

Saria Al Lahham

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SUMMARY

Senior AI/ML Engineer with more than 5 years of experience in Deep Learning, Generative AI, LLMs, and agentic systems. Currently serving as AI Technical Lead at Cisco working on agentic AI for large scale software systems. Before that, I served as the AI R&D Manager at Apertera, leading the design and deployment of agentic LLM systems, RAG pipelines, and MLOps workflows for production translations. Previously was an AI/ML Research Engineer at Samsung Research America (AI Center, Montreal), working on embedded AI, wireless communications (5G/6G), indoor localization, and human-state estimation.

EDUCATION

Queen's University

Kingston, ON, Canada

Ph.D. in Computer Science; GPA: 4.23/4.30

Thesis: Quantifying Computational Reliability in Extreme Edge Computing: Applications to Distributed Edge AI

Queen's University

Kingston, ON, Canada

M.Sc. in Computer Science; GPA: 4.30/4.30

Thesis: Multi-Orchestrator Mobile Edge Learning: Designing Energy-Efficient Task Allocation and Incentive Schemes

Qatar University

Doha, Qatar

B.Sc. in Computer Engineering; GPA: 3.90/4.00

Graduated with High Order of Excellence

Senior Project: Designing a Smart Home Controller for Smart Home Devices using Hand Gestures

JOB EXPERIENCE

Cisco Systems, Inc.

Montreal, QC, Canada

AI Technical Lead (Operations and Development)

Jul 2026 – Present

- Architects and deploys enterprise-grade agentic AI solutions integrating LLMs with networking and telemetry data to accelerate the software development lifecycle.
- Building automated root cause analysis (RCA) tooling and multi-agent architectures, delivering AI-driven insights that streamline debugging and validation across engineering teams.
- Integrating enterprise data sources with LLMs through intelligent retrieval, context management, and multi-step reasoning, with first-class tool calling and memory.
- Leading enterprise-scale deployment of ML and GenAI systems with strong reliability, observability, and low-latency guarantees in production.
- Partnering with research, data engineering, and product teams to translate business requirements into robust, production-ready AI solutions.

Apertera (a.k.a AlexaTranslations)

Montreal, QC, Canada

AI R&D Manager

Jul 2025 – Jun 2026

- Architected AI services across the product suite, defining system design and orchestration for production LLM and agentic workloads.
- Developed sovereign AI capabilities addressing PII, sensitive data classification, data residency, and legal constraints around model training, fine-tuning, and inference.
- Translated business problems into scalable AI solutions, partnering with cross-functional teams to align technical strategy with product and operational needs.
- Proposed and drove new AI initiatives, surfacing opportunities to Product and owning the full lifecycle from idea to client-facing product or feature.

AI Engineer

Jul 2024 – Jun 2025

- Designed and deployed a RAG-powered translation system with agentic LLM orchestration, outperforming commercial translation engines on in-house data.

- Designed and implemented automated metric-collection and quality-evaluation pipelines for continuous production monitoring.
- Evaluated recent GenAI and LLM methods, benchmarking against production systems to guide architecture decisions.
- Collaborated with Product and Linguistic Operations teams to deploy features, evaluate system outputs, and curate training data.

Samsung Research America, AI Center

Montreal, QC, Canada

AI/ML Research Engineer

Oct 2022 – Feb 2024

- Developed, deployed, and tested embedded AI algorithms on real hardware and simulations for indoor localization, computer vision, and RAN optimizations.
- Translated mathematical and algorithmic specifications into efficient deployable code.
- Authored scientific papers and patent applications.
- Collaborated with Samsung business units to identify and scope new AI-driven projects.

Qatar University, Smart IoT Systems Lab

Doha, Qatar

Research Assistant

May 2020 – Sep 2021

- Designed and implemented AI-based protocols for Ultra Reliable Low Latency Communication (URLLC) in smart health systems.
- Surveyed and integrated recent algorithms into protocol designs.
- Authored and reviewed research articles.

SKILLS & EXPERTISE

Software Development

- **Programming Languages:** Python, MATLAB, C/C++, Java, and Bash scripting.
- **Machine Learning:** SciPy, NumPy, SciKit-Learn, Pandas, XGBoost, Matplotlib and Seaborn.
- **Deep Learning:** PyTorch, Tensorflow/Keras, JAX, OpenCV, and Vision Transformers.
- **Generative AI:** SGLang, vLLM, RAG, HuggingFace, Unsloth, OpenAI/LLM APIs, LangChain/LangGraph.
- **MLOps:** LightningAI, Weights & Biases, Docker, AWS Cloud, OpenSearch and Azure Cloud.
- **Simulation & Digital Twins:** Nvidia Sionna, Blender

Artificial Intelligence

- Data Science, Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, Multimodal Foundation Models, RAG, Knowledge Graphs, Agentic AI, Reinforcement Learning, and Multi-Agent Systems.

Telecommunications and Computer Networks

- Digital Signal Processing, Edge Networks, IoT, TCP/IP, Wireless and Cellular Networks (5G/6G), 3GPP/IEEE/ETSI Standards, and Digital Twins.

PATENTS

US 2024/0422642 A1 — Probabilistic Mobility Load Balancing in Multi-Band Mobile Communication Network

Dec 19, 2024

U.S. Patent Application No. 18/593,262 (Published). Applicant: Samsung Electronics Co., Ltd

- **Inventors:** MHD Saria Al Lahham; Di Wu; Abu Zafar Ekram Hossain; Xue Liu; Gregory L. Dudek

LIST OF PUBLICATIONS

- [1] Alaa Awad Abdellatif, **Allahham, Mhd Saria**, Noor Khial, Amr Mohamed, Aiman Erbad, and Khaled Shaban. Reliable federated learning for age sensitive mobile edge computing systems. In *ICC 2023-IEEE International Conference on Communications*, pages 1622–1627. IEEE, 2023.
- [2] Alaa Awad Abdellatif, **Allahham, Mhd Saria**, Amr Mohamed, Aiman Erbad, and Mohsen Guizani. Onsr: An optimal network selection and resource allocation framework in multi-rat systems. In *ICC 2021-IEEE International Conference on Communications*, pages 1–6. IEEE, 2021.
- [3] Sherif B. Azmy, **Allahham, Mhd Saria**, Nizar Zorba, and Hossam S. Hassanein. Quantifying the impact of incentives on service availability at the extreme edge. In *GLOBECOM 2024 - 2024 IEEE Global Communications Conference*, pages 4162–4167, 2024.

- [4] Emna Baccour, **Allahham, Mhd Saria**, Aiman Erbad, Amr Mohamed, Ahmed Refaey Hussein, and Mounir Hamdi. Zero touch realization of pervasive artificial intelligence as a service in 6g networks. *IEEE Communications Magazine*, 61(2):110–116, 2023.
- [5] Yahuza Bello, Alaa Awad Abdellatif, **Allahham, Mhd Saria**, Ahmed Refaey Hussein, Aiman Erbad, Amr Mohamed, and Mohsen Guizani. B5g: Predictive container auto-scaling for cellular evolved packet core. *IEEE Access*, 9:158204–158214, 2021.
- [6] Heba DM Dawoud, **Allahham, Mhd Saria**, Alaa Awad Abdellatif, Amr Mohamed, Aiman Erbad, and Mohsen Guizani. Patient-driven network selection in multi-rat health systems using deep reinforcement learning. In *2021 IEEE Global Communications Conference (GLOBECOM)*, pages 1–6. IEEE, 2021.
- [7] Noor Khial, **Allahham, Mhd Saria**, Naram Mhaisen, Loay Ismail, Mohamed Mabrok, and Amr Mohamed. Drone-rl: Dynamic reinforcement learning for online navigation of uavs in evolving environments. *Knowledge-Based Systems*, page 115147, 2025.
- [8] Naram Mhaisen, **Allahham, Mhd Saria**, Amr Mohamed, Aiman Erbad, and Mohsen Guizani. On designing smart agents for service provisioning in blockchain-powered systems. *IEEE Transactions on Network Science and Engineering*, 9(2):401–415, 2021.
- [9] Hassan Saadat, **Allahham, Mhd Saria**, Alaa Awad Abdellatif, Aiman Erbad, and Amr Mohamed. RL-assisted energy-aware user-edge association for iot-based hierarchical federated learning. In *2022 International Wireless Communications and Mobile Computing (IWCMC)*, pages 548–553. IEEE, 2022.
- [10] **Al Laham, Saria**, Bobak H Baghi, Pierre-Yves Lajoie, Amal Feriani, Sachini Herath, Steve Liu, and Gregory Dudek. Device-free human state estimation using uwb multi-static radios. *arXiv preprint arXiv:2401.05410*, 2023.
- [11] **Allahham, Mhd Saria**, Alaa Awad Abdellatif, Naram Mhaisen, Amr Mohamed, Aiman Erbad, and Mohsen Guizani. Multi-agent reinforcement learning for network selection and resource allocation in heterogeneous multi-rat networks. *IEEE Transactions on Cognitive Communications and Networking*, 8(2):1287–1300, 2022.
- [12] **Allahham, Mhd Saria**, Alaa Awad Abdellatif, Amr Mohamed, Aiman Erbad, Elias Yaacoub, and Mohsen Guizani. I-see: Intelligent, secure, and energy-efficient techniques for medical data transmission using deep reinforcement learning. *IEEE Internet of Things Journal*, 8(8):6454–6468, 2020.
- [13] **Allahham, Mhd Saria**, Mohammad F Al-Sa’d, Abdulla Al-Ali, Amr Mohamed, Tamer Khattab, and Aiman Erbad. Dronerf dataset: A dataset of drones for rf-based detection, classification and identification. *Data in brief*, 26:104313, 2019.
- [14] **Allahham, Mhd Saria**, Salimur Choudhury, and Hossam S. Hassanein. Reliable federated learning with auction-based incentives at the extreme edge. In *GLOBECOM 2024 - 2024 IEEE Global Communications Conference*, pages 3134–3139, 2024.
- [15] **Allahham, Mhd Saria** and Hossam S. Hassanein. Quantifying computational reliability of task assignment in extreme edge computing. *IEEE Transactions on Mobile Computing*, pages 1–15, 2026.
- [16] **Allahham, Mhd Saria**, Tamer Khattab, and Amr Mohamed. Deep learning for rf-based drone detection and identification: A multi-channel 1-d convolutional neural networks approach. In *2020 IEEE International Conference on Informatics, IoT, and Enabling Technologies (ICIoT)*, pages 112–117. IEEE, 2020.
- [17] **Allahham, Mhd Saria**, Amr Mohamed, Aiman Erbad, and Mohsen Guizani. Motivating learners in multi-orchestrator mobile edge learning: A stackelberg game approach. *IEEE Canadian Journal of Electrical and Computer Engineering*, 46(1):69–76, 2022.
- [18] **Allahham, Mhd Saria**, Amr Mohamed, Aiman Erbad, and Hossam Hassanein. On the modeling of reliability in extreme edge computing systems. In *2022 5th International Conference on Communications, Signal Processing, and their Applications (ICCSPA)*, pages 1–6. IEEE, 2022.

- [19] **Allahham, Mhd Saria**, Amr Mohamed, and Hossam Hassanein. Incentive-based resource allocation for mobile edge learning. In *2022 IEEE 47th Conference on Local Computer Networks (LCN)*, pages 157–164. IEEE, 2022.
- [20] **Allahham, Mhd Saria**, Sameh Sorour, Amr Mohamed, Aiman Erbad, and Mohsen Guizani. Energy-efficient multi-orchestrator mobile edge learning. *arXiv preprint arXiv:2109.00757*, 2021.
- [21] **Lahham, Saria Al**, Di Wu, Ekram Hossain, Xue Liu, and Gregory Dudek. Probabilistic mobility load balancing for multi-band 5g and beyond networks. In *2024 IEEE International Conference on Communications Workshops (ICC Workshops)*, pages 1673–1678, 2024.